

A proof of Conjecture 00000004910

Merging duplicate actions: two MDPs with the same optimal policies and value functions but different transition structures

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1 The conjecture and its reading

Conjecture 00000004910 reads (English version; the Chinese version says the same):

Definition: The optimal policy and the transition structure are two layers of model data. *Conjecture:* There exist two MDPs with identical optimal policies and value functions but different transition structures, and the separation is realized by an explicit construction merging redundant actions.

Reading. An action is *redundant* when another action has the same transition law and the same reward at every state (an exact duplicate): deleting or identifying it loses nothing. *Merging* redundant actions identifies each class of duplicates into a single action, i.e. the action set is replaced by its quotient. So we start from an MDP M that genuinely has duplicate actions and let M' be the merged MDP. The two MDPs then have different action sets and different transition kernels, and a policy of M is not literally a policy of M' . “Identical optimal policies” is therefore read *modulo the identification of duplicate actions*: there is an explicit map q_* from policies of M onto policies of M' (forget which duplicate was played) such that

$$\text{Opt}(M) = q_*^{-1}(\text{Opt}(M')) \quad \text{and} \quad \text{Opt}(M') = q_*(\text{Opt}(M)),$$

where Opt is the set of optimal history-dependent randomized policies. Thus the two optimal sets correspond exactly: a policy is optimal in M iff its image is optimal in M' , and every optimal policy of M' is the image of an optimal policy of M . The value functions are literally equal. We prove this for every MDP with duplicate actions and every discount factor, and give an explicit two-state witness.

2 Definitions (as formalized)

An MDP on a state type S and an action type A is a pair (P, r) with transition laws $P(s, a) \in \text{PMF}(S)$ and rewards $r(s, a) \in [0, \infty)$ (Lean: **MDP** **S** **A**). A history is a list of past state-action pairs together with the current state; a policy π maps histories to probability distributions on A (history-dependent and randomized; **Policy** **S** **A**). For a start state s , the law $\mathcal{L}_{s,t}^\pi$ of the history at time t is defined by the usual recursion (**law**): $\mathcal{L}_{s,0}^\pi = \delta_{(\llbracket, s)}$, and a time- $(t+1)$ history is obtained by drawing $h \sim \mathcal{L}_{s,t}^\pi$, $a \sim \pi(h)$, $s' \sim P(\text{current state of } h, a)$ and appending. $E_s^\pi[r_t]$ is the expected reward at time t (**expReward**), and

$$v^\pi(s, \beta) = \sum_{t \geq 0} \beta^t E_s^\pi[r_t], \quad V^*(s, \beta) = \sup_{\pi} v^\pi(s, \beta)$$

(discValue, optValue). The *value function* is $s \mapsto V^*(s, \beta)$. A policy is *optimal* for β if $v^\pi(s, \beta) = V^*(s, \beta)$ for every state s (IsOptimal). This framework is adapted from the package for conjecture 00000004930.

Conventions. Values live in $[0, \infty]$ with truncated subtraction, so $(1 - \beta)^{-1}$ equals $1/(1 - \beta)$ for $\beta \in [0, 1)$ and ∞ for $\beta \geq 1$. All results below are stated for every $\beta \in [0, \infty]$; the meaningful range is $\beta \in [0, 1)$, including $\beta = 0$. (For $\beta \geq 1$ the witness has $V^* \equiv \infty$.)

3 Merging duplicate actions

Fix $q : A \rightarrow B$ and $g : B \rightarrow A$ with $q(g(b)) = b$, and assume every q -fibre consists of duplicates:

$$P(s, g(q(a))) = P(s, a), \quad r(s, g(q(a))) = r(s, a) \quad \text{for all } s, a. \quad (\text{D})$$

The *merged MDP* $M' = \text{quot } M \text{ } g$ has action type B , $P'(s, b) = P(s, g(b))$ and $r'(s, b) = r(s, g(b))$; by (D) this does not depend on the choice of the section g .

For a history h of M , $\bar{h} = \text{histMap } q \text{ } h$ replaces every past action a by $q(a)$. The *lift* of a policy π' of M' is $\text{lift } \pi'(h) = g_* \pi'(\bar{h})$ (look at the merged history, play the representative).

Pushforward (push). For a history h' of M' let $\text{start}(h')$ be its initial state and $|h'|$ its length (time). For a policy π of M , start state s and time t , let $J_{s,t}^\pi$ be the joint law of $(\bar{h}, q(a))$ where $h \sim \mathcal{L}_{s,t}^\pi$ and $a \sim \pi(h)$ (joint). Define

$$q_* \pi(h')(b) = \frac{J_{s,t}^\pi(h', b)}{\sum_{b'} J_{s,t}^\pi(h', b')} \quad (s = \text{start}(h'), t = |h'|),$$

the conditional law of the merged action given the merged history, when the denominator is nonzero; otherwise $q_* \pi(h') = q_*(\pi(\text{the history } h' \text{ with each } b \text{ replaced by } g(b)))$. This is a history-dependent randomized policy of M' .

Lemma 1 (law_support, joint_eq). *Every history in the support of $\mathcal{L}_{s,t}^\pi$ has start state s and length t . Hence $J_{s,t}^\pi$ disintegrates as “draw h' from the merged history law $q_\# \mathcal{L}_{s,t}^\pi$, then $b \sim q_* \pi(h')$ ”: $J_{s,t}^\pi(h', b) = q_\# \mathcal{L}_{s,t}^\pi(h') \cdot q_* \pi(h')(b)$.*

Proof. The support claim is an induction on t . The first marginal of $J_{s,t}^\pi$ is $q_\# \mathcal{L}_{s,t}^\pi$ (joint_fst, tsum_pair). If $m := q_\# \mathcal{L}_{s,t}^\pi(h') = 0$ both sides vanish. Otherwise $h' = \bar{h}$ for some h in the support, so $\text{start}(h') = s$ and $|h'| = t$; thus $q_* \pi(h')$ is computed from $J_{s,t}^\pi$ itself, the denominator equals $m \in (0, 1]$, and $m \cdot J(h', b)/m = J(h', b)$. $\square$

Lemma 2 (law_push, discValue_push). *Under (D), for every policy π of M , start s and time t : $\mathcal{L}_{s,t}^{q_* \pi}(M') = q_\# \mathcal{L}_{s,t}^\pi(M)$, i.e. the merged history process of π (all states, merged actions) has exactly the law of the history process of $q_* \pi$ in M' . Consequently $v_{M'}^{q_* \pi}(s, \beta) = v_M^\pi(s, \beta)$ for all s, β .*

Proof. Induction on t ; $t = 0$ is immediate. For the step, by the induction hypothesis and Lemma 1 the time- $(t + 1)$ law in M' is obtained by drawing $(h', b) \sim J_{s,t}^\pi$ and then $s' \sim P'(\cdot, b)$. Unfolding J , this is: draw $h \sim \mathcal{L}_{s,t}^\pi$, $a \sim \pi(h)$, $s' \sim P(\cdot, g(q(a))) = P(\cdot, a)$ by (D), and append $(\cdot, q(a))$ to $\bar{h}$ — which is the image under $h \mapsto \bar{h}$ of the time- $(t + 1)$ law in M . For values, $E_s^{q_* \pi}[r_t] = \sum_{(h', b)} J_{s,t}^\pi(h', b) r(\cdot, g(b))$ by Lemma 1, and unfolding J and using (D) for r this is $E_s^\pi[r_t]$. $\square$

Lemma 3 (law_lift, discValue_lift, push_lift). *For every policy π' of M' : the merged history law of $\text{lift } \pi'$ in M is the history law of π' in M' , so $v_M^{\text{lift } \pi'} = v_{M'}^{\pi'}$; and $q_*(\text{lift } \pi') = \pi'$. In particular q_* is onto.*

Proof. The first two claims are an induction as before (only $q \circ g = \text{id}$ is used). For the last, $J_{s,t}^{\text{lift}} \pi'(h', b) = m(h') \pi'(h')(b)$ with $m = q_{\#} \mathcal{L}_{s,t}^{\text{lift}} \pi'$; comparing with Lemma 1 gives $q_*(\text{lift } \pi')(h') = \pi'(h')$ whenever $m(h') \neq 0$. If $m(h') = 0$ the fallback branch gives $q_* g_* \pi'(h') = \pi'(h')$. $\square$

Theorem 4 (`optValue_quot`, `isOptimal_push`, `isOptimal_lift`). *Under (D), for every $\beta \in [0, \infty]$: $V_{M'}^*(s, \beta) = V_M^*(s, \beta)$ for every s ; π is optimal in M iff $q_* \pi$ is optimal in M' ; π' is optimal in M' iff $\text{lift } \pi'$ is optimal in M . Hence $\text{Opt}(M) = q_*^{-1}(\text{Opt}(M'))$ and $\text{Opt}(M') = q_*(\text{Opt}(M))$.*

Proof. Every value of M' is a value of M (Lemma 3) and every value of M is a value of M' (Lemma 2), so the suprema agree; the optimality equivalences follow from the value identities. If $\pi' \in \text{Opt}(M')$ then $\text{lift } \pi' \in \text{Opt}(M)$ and $q_*(\text{lift } \pi') = \pi'$. $\square$

Note that q_* is many-to-one: optimal policies of M that play red_2 (below) at some histories, reachable or not, are sent to the same optimal policy of M' as policies playing red_1 there. This is exactly the identification of duplicates.

4 The witness

States $\{\text{false}, \text{true}\}$. M_2 has actions $\{\text{opt}, \text{red}_1, \text{red}_2\}$: opt has reward 1 and stays at the current state; red_1 and red_2 have reward 0 and both move to the other state. So $\text{red}_1, \text{red}_2$ are exact duplicates. The merge map q sends $\text{opt} \mapsto \text{opt}$ and $\text{red}_1, \text{red}_2 \mapsto \text{red}$, with section $\text{rep}(\text{red}) = \text{red}_1$, and $M_3 = \text{quot } M_2 \text{ rep}$ has actions $\{\text{opt}, \text{red}\}$. (D) holds by inspection.

Different transition structures. M_2 has three actions and $a \mapsto P_2(s, a)$ is not injective at any state ($P_2(s, \text{red}_1) = P_2(s, \text{red}_2)$); M_3 has two actions and $b \mapsto P_3(s, b)$ is injective at every state ($\delta_s \neq \delta_{-s}$).

Lemma 5 (`optValue_eq`, `constPol_optimal`). *If $r \leq 1$ and some action o has $r(s, o) = 1$ at every state, then $V^*(s, \beta) = (1 - \beta)^{-1}$ for every s and β , and the policy always playing o is optimal.*

Proof. $E_s^\pi[r_t] \leq 1$, so $v^\pi(s, \beta) \leq \sum_t \beta^t = (1 - \beta)^{-1}$, attained by always playing o . $\square$

Theorem 6 (`conjecture4910`). *M_2 has the duplicate actions $\text{red}_1, \text{red}_2$, $M_3 = \text{quot } M_2 \text{ rep}$ merges them, $q \circ \text{rep} = \text{id}$, $P_2(s, \cdot)$ is not injective and $P_3(s, \cdot)$ is injective for every s . For every s, t, β and policies π of M_2 , π' of M_3 :*

1. $V_{M_2}^*(s, \beta) = V_{M_3}^*(s, \beta) = (1 - \beta)^{-1}$;
2. $\mathcal{L}_{s,t}^{q_* \pi}(M_3) = q_{\#} \mathcal{L}_{s,t}^\pi(M_2)$ and $v_{M_3}^{q_* \pi}(s, \beta) = v_{M_2}^\pi(s, \beta)$; $q_*(\text{lift } \pi') = \pi'$;
3. $\text{Opt}_\beta(M_2) = q_*^{-1}(\text{Opt}_\beta(M_3))$ and $\text{Opt}_\beta(M_3) = q_*(\text{Opt}_\beta(M_2))$;
4. the policy always playing opt is optimal in both, so both sets are nonempty.

Proof. Theorem 4 with $M = M_2$, and the previous lemma with $o = \text{opt}$. $\square$

Remark 7. The witness is deliberately small (rewards depend only on the action, so V^* is constant); the general Theorem 4 holds for every MDP with duplicate actions. An earlier version of this package also treated a reading in which a dominated action is redirected to another action's law within the same action set; that reading has been dropped, since the redirected actions were not duplicates beforehand.

5 Lean formalization

File `lean/Conjecture4910/Basic.lean` (namespace `Conjecture4910`), Lean 4.33.1 with Mathlib v4.33.1, 370 lines. Main theorem `Conjecture4910.conjecture4910`; general results `joint_eq`, `law_push`, `discValue_push`, `law_lift`, `discValue_lift`, `push_lift`, `optValue_quot`, `isOptimal_push`, `isOptimal_lift`, `optValue_eq`. `#print axioms Conjecture4910.conjecture4910` reports only `propext`, `Classical.choice`, `Quot.sound`. There is no `sorry` and no `native_decide`. The MDP framework (`structure`, `histories`, `policies`, `law`, `expReward`, `discValue`, `optValue`) is adapted from the package for conjecture 00000004930, with the start state made a parameter.